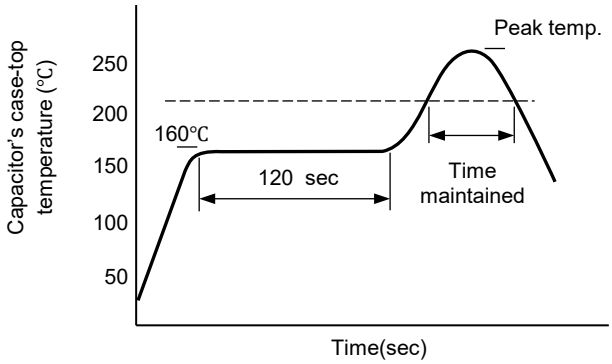
Recommended reflow soldering



Size code	C, D, D8	F, G, G12, G16	
Peak temp.	260℃ (255℃)	245℃	260℃
Time in peak temperature	≧ 250℃ 5 s (10 s)	≧ 240℃ 10 s	≧ 250℃ 5 s
Time maintained	≧ 230℃ 30 s	≧ 230℃ 30 s	≧ 230℃ 30 s
	≧ 217℃ 40 s	≧ 217℃ 40 s	≧ 217℃ 40 s
	≧ 200℃ 70 s	≧ 200℃ 70 s	≧ 200℃ 70 s
Reflow cycles	2 times	2 times	1 time

* For reflow, use a thermal condition system such as infrared and radiation (IR) or hot blas.
* Reflow temperature is measured on capacitor's case top.

Specifications for radial lead type

● Flow soldering condition

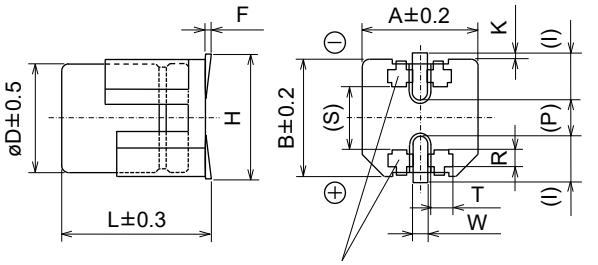
< RoHS compliant >

	Temperature	Time	Flow number
Soldering condition	260℃ + 5℃ or less	10 sec +1 sec or less	1 time

Dimensions (Vibration-proof products)

The size and shape are different frome standard products.

< Size code : D, D8 >



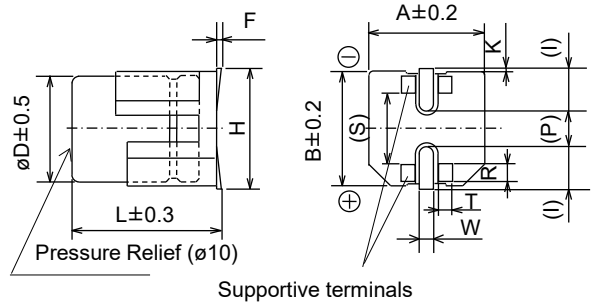
( ) Reference size

Unit : mm

Size code	øD	L	A, B	H max.	F	I	W
D	6.3	6.1	6.6	7.8	0 to +0.15	2.4	0.65±0.1
D8	6.3	8.0	6.6	7.8	0 to +0.15	2.4	0.65±0.1

Size code	P	K	R	S	T
D	2.2	0.35 $^{+0.15}_{-0.20}$	1.1±0.2	3.3	1.05±0.2
D8	2.2	0.35 $^{+0.15}_{-0.20}$	1.1±0.2	3.3	1.05±0.2

< Size code : F, G, G12, G16 >



( )Reference size

Unit : mm

Size code	øD	L	A, B	H max.	F	I	W
F	8.0	10.5	8.3	10.0	0 to +0.15	3.4	1.2±0.2
G	10.0	10.5	10.3	12.0	0 to +0.15	3.5	1.2±0.2
G12	10.0	12.8	10.3	11.0*1	0 to +0.15	3.2	1.2±0.2
G16	10.0	16.8	10.3	11.0*1	0 to +0.15	3.2	1.2±0.2

*1:±0.2

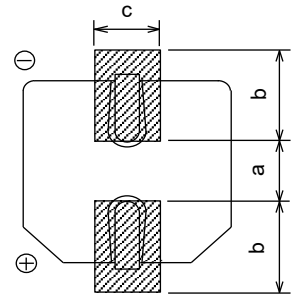
Size code	P	K	R	S	T
F	3.1	0.70±0.2	0.70±0.2	5.3	1.3±0.2
G	4.6	0.70±0.2	0.70±0.2	6.9	1.3±0.2
G12	4.6	–	0.70±0.2	6.9	1.3±0.2
G16	4.6	–	0.70±0.2	6.9	1.3±0.2

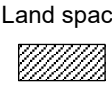
Mounting specification

Land / Pad pattern

The circuit board land/pad pattern size for chip capacitors is specified in the following table.
The land pitch influences installation strength.

● Standard products





Unit : mm

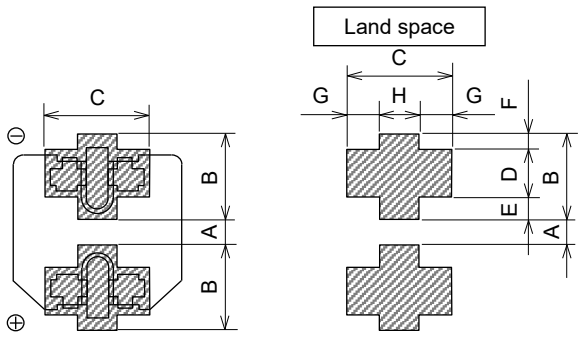
Size code	a	b	c
C : ø5×L5.8	1.5	2.8	1.6
D : ø6.3×L5.8	1.8	3.2	1.6
D8 : ø6.3×L7.7	1.8	3.2	1.6
F : ø8×L10.2	3.1	4.0	2.0
G : ø10×L10.2	4.6	4.1	2.0
G12 : ø10×L12.5	4.6	4.1	2.0
G16 : ø10×L16.5	4.6	4.1	2.0

Larger dimension of “a” may prevent back fillet from being formed adequately to obtain required solder strength.

＊Take mounting conditions, solderability and fitting strength into consideration when selecting parts for your design.

● Vibration-proof products

< Size code：D, D8 >



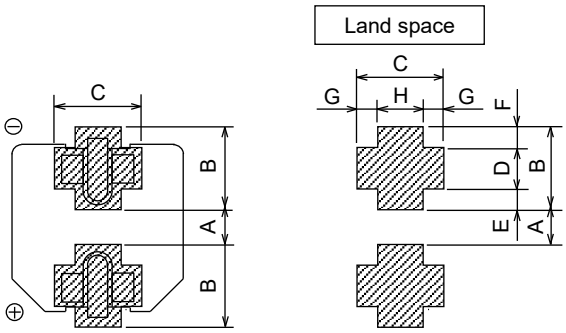
Unit : mm

Size code	A	B	C	D
D : ø6.3×L6.1	1.2	3.6	3.2	2.0
D8 : ø6.3×L8.0	1.2	3.6	3.2	2.0

Size code	E	F	G	H
D : ø6.3×L6.1	0.95	0.65	1.0	1.2
D8 : ø6.3×L8.0	0.95	0.65	1.0	1.2

Larger dimension of “A” may prevent back fillet from being formed adequately to obtain required solder strength.

< Size code：F, G, G12, G16 >



Unit : mm

Size code	A	B	C	D
F : ø8×L10.5	2.7	4.0	4.7	1.3
G : ø10×L10.5	3.9	4.4	4.7	1.3
G12 : ø10×L12.8	3.9	4.4	4.7	1.3
G16 : ø10×L16.8	3.9	4.4	4.7	1.3

Size code	E	F	G	H
F : ø8×L10.5	1.0	1.7	1.1	2.5
G : ø10×L10.5	1.2	1.9	1.1	2.5
G12 : ø10×L12.8	1.2	1.9	1.1	2.5
G16 : ø10×L16.8	1.2	1.9	1.1	2.5

Larger dimension of “A” may prevent back fillet from being formed adequately to obtain required solder strength.

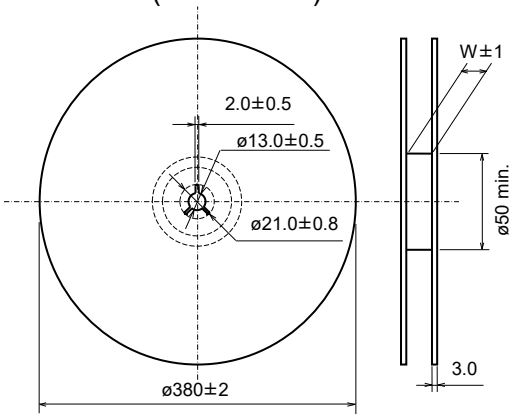
＊Take mounting conditions, solderability and fitting strength into consideration when selecting parts for your design.

＊ The vibration-proof capacitors of size ø6.3 has support terminals extending from the bottom side to the lead edge. Then, make sure to find appropriate soldering conditions to form fillet on the support terminals if required for appearance inspection.

Packaging specifications

Specifications for surface mount type

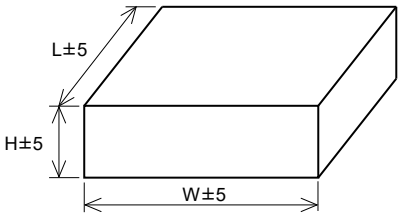
● Reel dimensions (not to scale)



Unit : mm

Size code	W
C	14.0
D, D8	18.0
F, G, G12, G16	26.0

● Dimensions of outer carton box



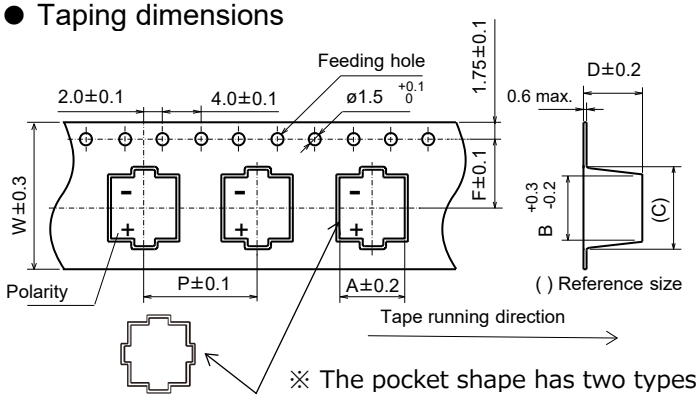
Unit : mm

Size code	H	W, L
C	180	395
D, D8	220	395
F, G, G12, G16	180	395

● Min.packing quantity

Size code	Min.packing quantity (pcs.)
C, D	1000
D8	900
F, G	500
G12	400
G16	250

● Taping dimensions



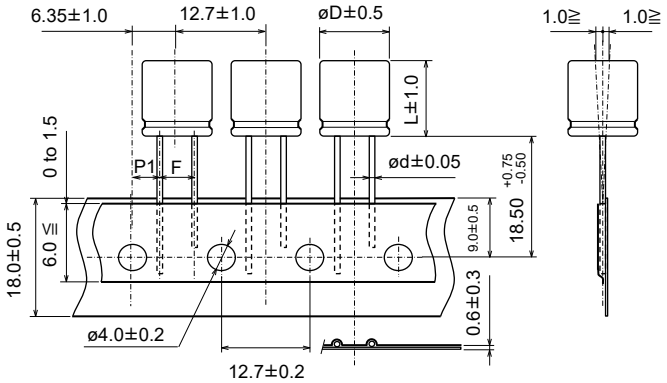
※ The pocket shape has two types depending on the size code.

Unit : mm

Size code	A	B	C	D	P	F	W
C	5.7	5.7	8.0	6.4	12.0	5.5	12.0
D	7.0	7.0	9.0	6.4	12.0	7.5	16.0
D8	7.0	7.0	9.0	8.4	12.0	7.5	16.0
F	8.7	8.7	12.5	11.0	16.0	11.5	24.0
G	10.7	10.7	14.5	11.0	16.0	11.5	24.0
G12	10.7	10.7	14.5	13.7	16.0	11.5	24.0
G16	10.7	10.7	14.5	17.5	20.0	11.5	24.0

Radial lead type

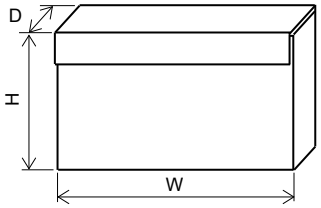
● Taping dimensions

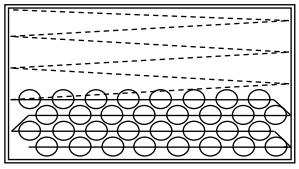


Unit : mm

Size code	øD	L	ød	F	P1
F	8.0	9.5	0.6	3.5±0.5	4.60±0.50
G	10.0	9.5	0.6	5.0+0.8/-0.2	3.85±0.50
G12	10.0	11.7	0.8	5.0+0.8/-0.2	3.85±0.50
G16	10.0	15.7	0.8	5.0+0.8/-0.2	3.85±0.50

● Dimensions of outer carton box / Packaging method





Zigzag folded

Unit : mm

Size code	W	H	D
F	340 max.	230 max.	55 max.
G	340 max.	170 max.	55 max.
G12	340 max.	170 max.	55 max.
G16	340 max.	170 max.	55 max.

● Min.packing quantity

Size code	Min.packing quantity (pcs.)
F	1000
G	500
G12	500
G16	500